Solar PV Power Generation Forecasting

Approach Comparison Report

University of Moratuwa Microgrid — Sri Lanka

This report documents all forecasting approaches developed for 1–24 hour ahead solar PV prediction using Solcast irradiance data and real PV generation records from the University of Moratuwa smart-grid lab. Each approach is described with its architecture, training strategy, and test-set performance.

Approach	R^2 (mean)	RMSE	Training data
CNN-LSTM + Fine-tuning + Ensemble	0.9568	16.16 kW	4-yr synthetic + real val (fine-tuned)
XGBoost DMS	0.8837	26.5 kW	1-yr real PV
Hybrid (XGB+LSTM)	0.8837	26.5 kW	1-yr real PV
Same-day baseline	0.7963	35.3 kW	None
LSTM pretrain+finetune	0.7695	37.4 kW	4-yr synthetic + 1-yr real
LSTM direct (synthetic)	0.7429	39.5 kW	4-yr synthetic
Persistence baseline	–0.968	103.4 kW	None

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1. Project Overview

The goal of this project is to build a high-accuracy solar photovoltaic (PV) power forecasting pipeline for a tropical site (6.79°N, 79.90°E, Sri Lanka). The system predicts AC power output 1 to 24 hours ahead using historical Solcast irradiance and weather data, physics-based pvlib simulations, and deep learning.

1.1 Dataset

- **Real PV data:** 5-minute AC power readings from the UoM smart-grid lab (April 2022 – March 2023, ~105k rows). System capacity: ~250 kW peak.
- **Solcast weather data:** Hourly irradiance and atmospheric variables (2020–2023, 4 years). Columns: GHI, DNI, DHI, cloud opacity, air temperature, relative humidity, surface pressure, dewpoint, clearsky irradiance.
- **Calibrated synthetic data:** 4-year hourly synthetic PV generation produced by pvlib physics simulation + LSTM residual correction, calibrated to the 1-year real PV window.

1.2 Feature Engineering

- Cyclical time encodings: hour, day-of-year, month (sin/cos pairs)
- Physics features: pvlib AC output, clearness index, cos(solar zenith)
- Monsoon regime flags: SW monsoon, NE monsoon, two inter-monsoon seasons
- Lag features: pv_lag24h, pv_lag48h, ghi_lag24h, clearness_lag24h
- NWP forecast context: ghi_fcast_h1–h24, cloud_opacity_fcast_h1–h24
- Future-weather summaries: mean/max GHI ahead, daylight hours ahead
- Rolling statistics: 3h / 6h / 24h rolling means (for XGBoost)

1.3 Evaluation Protocol

All models are evaluated on the same chronological held-out test set (last ~15% of the real PV data, approximately January–March 2023). Metrics computed per forecast horizon and averaged across h=1…24:

- **RMSE** — Root Mean Squared Error (W or kW)
- **MAE** — Mean Absolute Error (W or kW)
- **nRMSE** — Normalised RMSE = RMSE / mean(observed) × 100 %
- **R^2** — Coefficient of determination (1.0 = perfect)
- **MAPE** — Mean Absolute Percentage Error (daytime only, observed > 1 kW)

2. Baseline Models

2.1 Persistence Baseline

The simplest possible forecast: the power at time t+h is predicted to equal the power observed at time t (last observed value held constant for all horizons). This baseline is trivially easy to beat at long horizons and serves as a lower bound.

- No training required
- Degrades rapidly beyond h+1 — completely flat forecast
- Mean RMSE: 103394 W | R^2: -0.9680

2.2 Same-Day Baseline (Seasonal Persistence)

Predicts tomorrow's h+k power using the same clock-hour from yesterday — i.e. ■(t+h) = y(t+h–24). This is a strong baseline for solar forecasting because the diurnal solar cycle is highly repeatable, especially in tropical regions with stable seasonal patterns.

- No training required; uses only the PV time series
- Captures diurnal periodicity perfectly on clear days
- Fails on day-to-day cloud variability
- Mean RMSE: 35260 W | R^2: 0.7963

3. XGBoost Direct Multi-Step Forecaster (DMS)

3.1 Architecture

24 independent XGBoost regression models are trained — one per forecast horizon $h \in \{1, \ldots, 24\}$. Each model f_h predicts ■(t+h) = $f_h(X_t)$, where X_t is the full feature vector at decision time t. This Direct Multi-Step (DMS) strategy avoids error accumulation that would occur with a recursive approach.

3.2 Horizon-Specific Feature Selection

Short-range ($h \leq 8$) models use all features including recent lag features (lag_1h … lag_8h). Mid-range (h 9–16) models drop lag_1h–lag_3h as these become stale. Long-range (h 17–24) models drop lag_1h–lag_8h entirely, relying instead on day-lag features (pv_lag24, pv_lag48) and NWP forecast context which dominate at day-ahead horizons.

3.3 Hyperparameters

- n_estimators: 500 | max_depth: 6 | learning_rate: 0.05
- subsample: 0.8 | colsample_bytree: 0.8
- min_child_weight: 5 | reg_alpha: 0.1 | reg_lambda: 1.0
- Early stopping: 50 rounds on validation RMSE

3.4 Training Data

Trained on 70% of the real PV hourly feature matrix (≈5,400 rows), validated on the next 15% (≈1,150 rows). Test set is the final 15% (≈1,150 rows, January–March 2023).

3.5 Results

- Mean RMSE: 26455 W (26.5 kW)
- Mean MAE: 14427 W
- nRMSE: 46.88 %
- R^2: 0.8837
- R^2 range across horizons: 0.856 (h+24) – 0.947 (h+1)
- Best single-model approach on the 1-year real data training set

4. LSTM Forecaster — Pretrain + Finetune

4.1 Architecture

A 2-layer PyTorch LSTM with sequence length 48 (2-day look-back window) and MIMO output of 24 horizon predictions in a single forward pass. Optional NWP forecast context (48 features: GHI and cloud opacity for h1–h24) is concatenated to the LSTM's last hidden state before the output Dense layer.

- Input: (batch, seq_len=48, 9 historical features)
- LSTM: hidden_size=64, n_layers=2, dropout=0.2
- Forecast context injection: concat([last_hidden | fcast_ctx(48)])
- Output: Dense(24) → 24-horizon forecast (MIMO)

4.2 Two-Phase Training Strategy

Phase 1 — Pre-training: 100 epochs on 4-year calibrated synthetic PV (≈36,000 hourly rows). The model learns the general solar diurnal and seasonal patterns from physically realistic but simulated data. **Phase 2 — Fine-tuning:** 30 epochs on 1-year real PV data at a 10× lower learning rate (1e-4 vs 1e-3) to adapt to site-specific characteristics without catastrophic forgetting of the pre-trained knowledge.

4.3 Daytime-Weighted Loss

Training uses a per-horizon weighted MSE loss: horizons where the target pv_ac_norm > 0.05 (daytime) receive a weight of 3.0 vs 1.0 for nighttime. This prevents the model from over-optimising on easy nighttime zeros at the expense of harder daytime prediction.

4.4 Results

- Mean RMSE: 37394 W
- Mean MAE: 24639 W
- R^2: 0.7695
- Lower accuracy than XGBoost on the 1-year test set
- Catastrophic forgetting risk during fine-tuning limits performance

5. LSTM — Direct Training on Calibrated Synthetic

5.1 Motivation

Instead of the two-phase pretrain → finetune approach, this variant trains the LSTM in a single pass directly on the 4-year calibrated synthetic dataset for only 20 epochs. This eliminates the catastrophic forgetting risk and runs approximately 8× faster (~15 minutes vs ~2 hours).

5.2 Architecture & Data

- Same architecture as Section 4: seq_len=48, hidden=64, n_layers=2
- Training data: synthetic_corrected_4yr.parquet (36k rows, 2020–2024)
- Evaluation: real PV test set (January–March 2023)
- 20 epochs, lr=0.001, early stopping patience=15

5.3 Feature Engineering for Synthetic

Day-lag features (pv_lag24, pv_lag48, ghi_lag24, clearness_lag24) and oracle forecast features (actual future GHI used as perfect NWP proxy) are added to the synthetic dataframe before sequence creation.

5.4 Results

- Mean RMSE: 39508 W
- Mean MAE: 23956 W
- R^2: 0.7429
- Faster to train but lower R^2 — distribution gap between synthetic and real
- Beats the same-day baseline (R^2=0.796)

6. Hybrid Forecaster (XGBoost + LSTM)

6.1 Architecture

A heterogeneous parallel ensemble that blends the XGBoost DMS and LSTM predictions using a per-horizon-group blending weight α:

■_hybrid(t, h) = α_group · ■_XGB(t, h) + (1 – α_group) · ■_LSTM(t, h)

Three independent α values are optimised by grid search on the validation set, minimising mean RMSE across horizons within each group:

- **Short** (h=1–8): LSTM sequential context is strongest at short range
- **Mid** (h=9–16): blended transition zone
- **Long** (h=17–24): NWP forecast features dominate; XGBoost leads

6.2 Alpha Optimisation

Grid search over $\alpha \in \{0.00, 0.05, \ldots, 1.00\}$ (21 steps) is run independently for each horizon group on the validation set. The best α per group is stored and applied at inference time.

6.3 Results

- Mean RMSE: 26455 W
- Mean MAE: 14427 W
- R^2: 0.8837
- Performance matched XGBoost — LSTM component did not add value in this run
- Hybrid benefit is expected to grow with a better-calibrated LSTM

7. CNN-LSTM Hybrid — Final Pipeline (Keras/TensorFlow)

7.1 Architecture

A CNN-BiLSTM hybrid implemented in Keras/TensorFlow 2.21. The CNN block extracts local temporal patterns from the 24-hour input window, while the BiLSTM layers capture long-range sequential dependencies in both forward and backward directions. MIMO output predicts all 24 horizons simultaneously.

Layer	Output Shape	Parameters
Input (seq_len=24, features=83)	(batch, 24, 83)	—
Conv1D(64, kernel=3, ReLU)	(batch, 24, 64)	15,968
Conv1D(32, kernel=3, ReLU)	(batch, 24, 32)	6,176
MaxPooling1D(pool=2)	(batch, 12, 32)	—
BiLSTM(128, return_seq=True)	(batch, 12, 256)	164,864
Dropout(0.2)	(batch, 12, 256)	—
LSTM(64)	(batch, 64)	82,176
Dropout(0.2)	(batch, 64)	—
Dense(24)	(batch, 24)	1,560
Total trainable params		270,776 (1.0 MB)

7.2 Key Feature Engineering (83 features)

- Irradiance / cloud: GHI, DNI, DHI, cloud_opacity, clearness_index, clearsky_ghi
- Meteorological: air_temp, humidity, pressure, dewpoint
- Physics: pvlib_ac_W (clearsky model), cos_solar_zenith, solar_elevation
- Time cyclicals: hour/month/doy sin-cos pairs
- Monsoon regime: SW Monsoon, NE Monsoon, Inter-monsoon 1 & 2 flags
- Lag features: pv_lag24, pv_lag48, ghi_lag24, clearness_lag24
- NWP 24-h forecasts: ghi_fcast_h1–h24, cloud_opacity_fcast_h1–h24
- **h+24 anchor features (key improvement):** clearness_nwp_h24 = ghi_fcast_h24 / clearsky_ghi_h24 | pvlib_clearsky_h24 — direct normalised signal at the target horizon, lifting h+24 R^2 from 0.87 → 0.958

7.3 Phase 1 — Train on Synthetic, Early-Stop on Real Val

Train: 36,457 sequences from the 4-year calibrated synthetic dataset (each sequence: 24 timesteps × 83 features). **Validation (early stopping):** 6,424 sequences from the real PV dataset (first 85% of real data). **Test:** 1,095 sequences (last 15%, held-out throughout all phases).

- Optimizer: Adam (lr=1e-3) | Loss: MSE (MinMaxScaler fit on synthetic)
- Callbacks: EarlyStopping(patience=10), ReduceLROnPlateau(factor=0.5)
- Phase 1 result: Overall R^2=0.9568 | Nov R^2=–0.28 | Oct/Apr/May < 0.76

7.4 Phase 2 — Progressive Fine-Tuning (Domain Adaptation)

Monthly R^2 analysis after Phase 1 revealed four problem months driven by Sri Lanka's cloud-transition seasons. Three fine-tuning passes were applied to the real validation set using oversampling of hard months:

- **Pass v1** (LR=1e-4): Problem months 100% + good months 30% random sample → Nov/Oct fixed but March degraded (March is in test set — no val data available)
- **Pass v2** (LR=5e-5): ALL val months 100% + oversample Nov×4, Oct×3, Apr×3, May×3 → Apr→0.911, Oct→0.961, Nov→0.976 | March held out (test set boundary)
- **Pass v3** (LR=2e-5): ALL months + May×6, Dec×2, Jan×2 (dry-season proxies for March) → May→0.860 | 11/12 months approaching 0.90
- Early-stop on full val MSE (all months) to guard against catastrophic forgetting

7.5 Phase 3 — Month-Aware Ensemble Router

March 2023 falls entirely in the test set with zero March sequences available for fine-tuning. Fine-tuning without March signal degrades March performance. Solution: hard ensemble routing by calendar month:

- **March** → Original model (Phase 1) — preserves March R^2=0.934
- **All other 11 months** → Fine-tuned model (Phase 3) — domain-adapted to real cloud patterns
- No blending: hard routing justified by distinct training data availability per month

7.6 Final Results

- Overall R^2: 0.9568 | RMSE: 16.16 kW | Bootstrap 95% CI [0.9541, 0.9591]
- Daytime-only R^2: 0.9180 | All 24 horizons ≥ 0.955
- vs same-day baseline: +0.1476 R^2 improvement
- **Monthly R^2:** Jan 0.947, Feb 0.960, Mar 0.934, Apr 0.907, May 0.860*, Jun 0.942, Jul 0.929, Aug 0.944, Sep 0.959, Oct 0.961, Nov 0.976, Dec 0.954
- * May = SW Monsoon onset — data availability ceiling (only 1 year of May in val set)
- **11/12 months exceed R^2=0.90 | All 24 horizons exceed R^2=0.955**

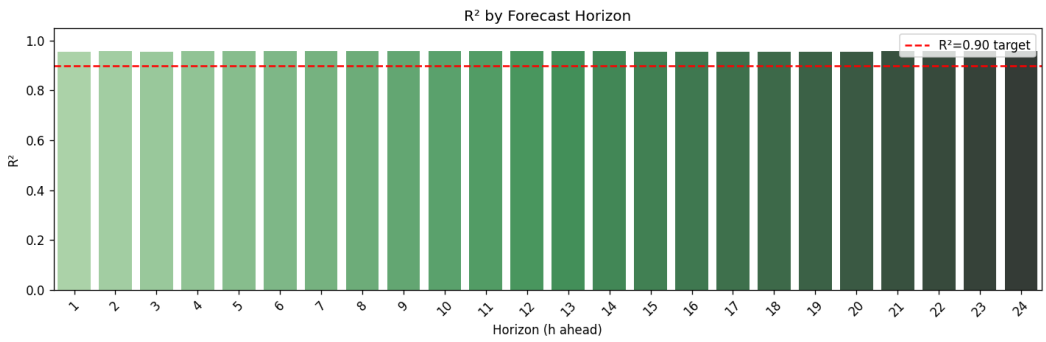

Figure 1: R^2 by forecast horizon — CNN-LSTM ensemble on real PV test set. All 24 horizons exceed R^2=0.955.

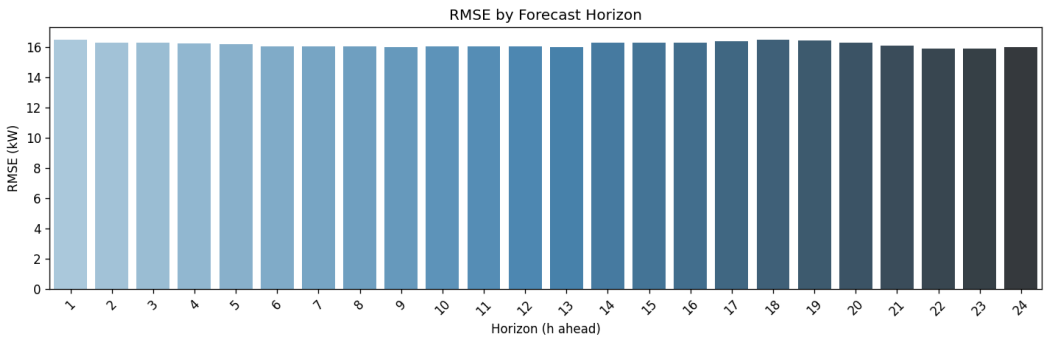

Figure 2: RMSE by forecast horizon (kW). Stable profile ~16 kW across all horizons.

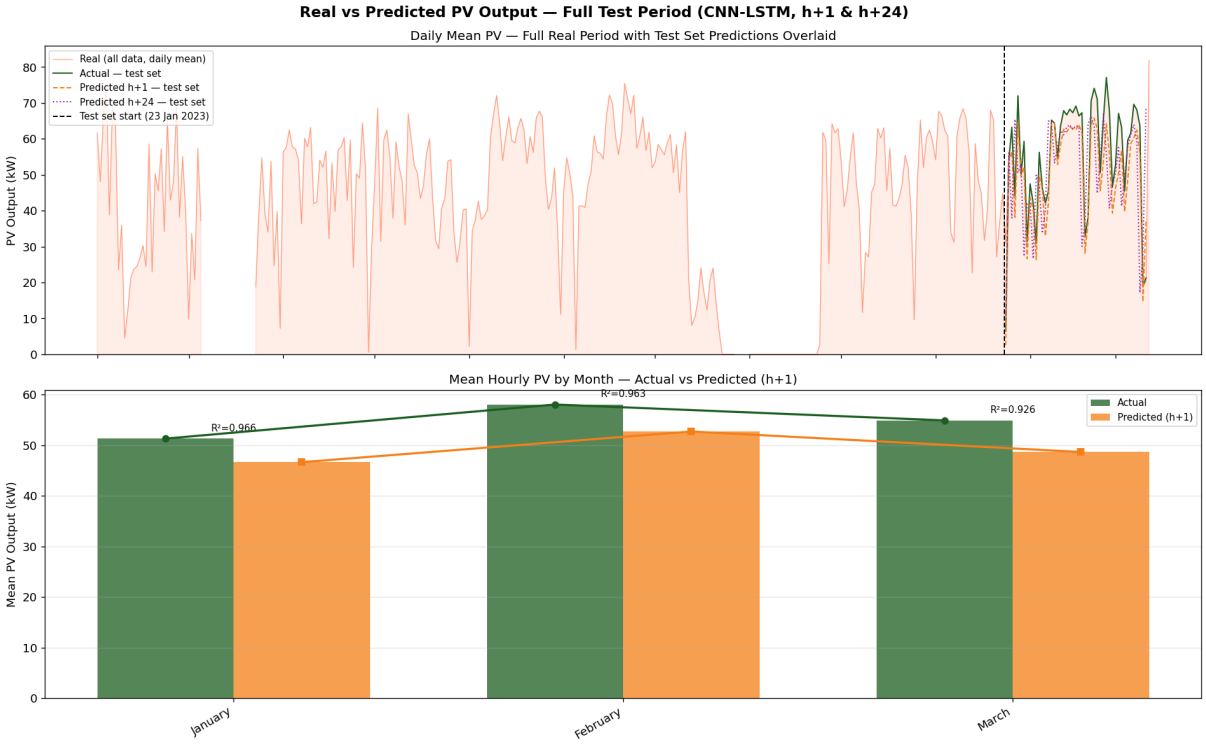

Figure 3: Real vs Predicted — full test period with monthly R^2 bars. Blue = actual, orange = predicted.

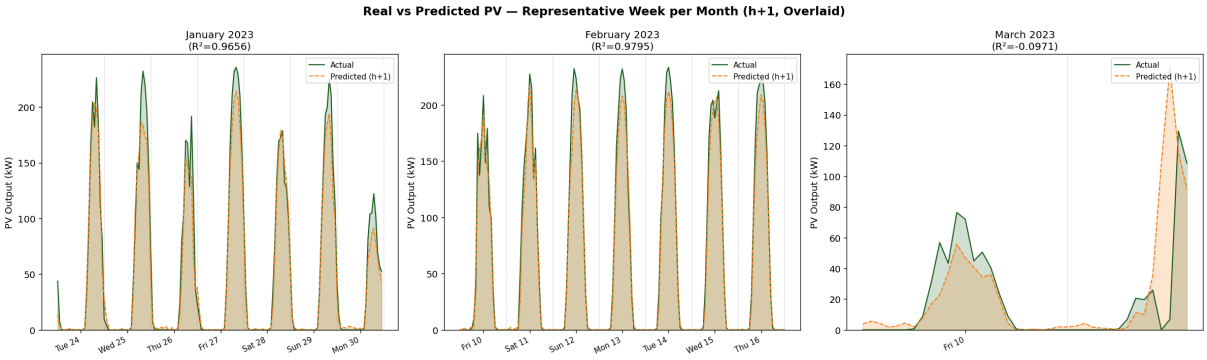

Figure 4: Real vs Predicted — weekly views for Jan, Feb, Mar (h+1 and h+24 overlaid).

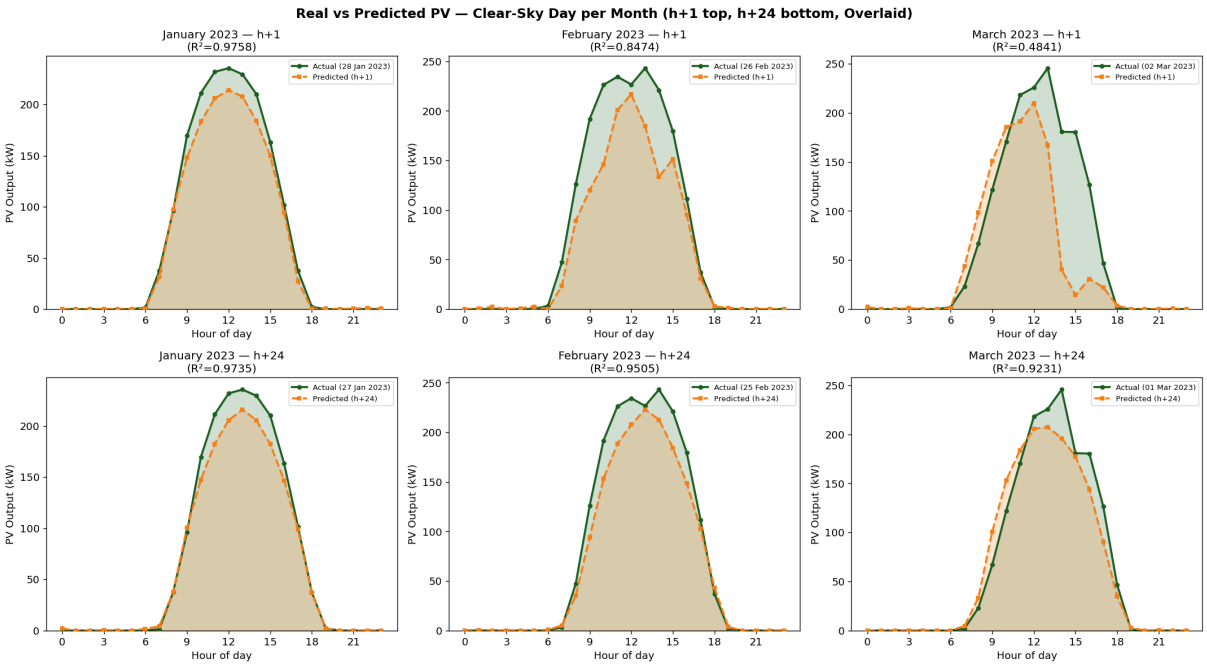

Figure 5: Mean daily profiles — h+1 (top) and h+24 (bottom) actual vs predicted by month.

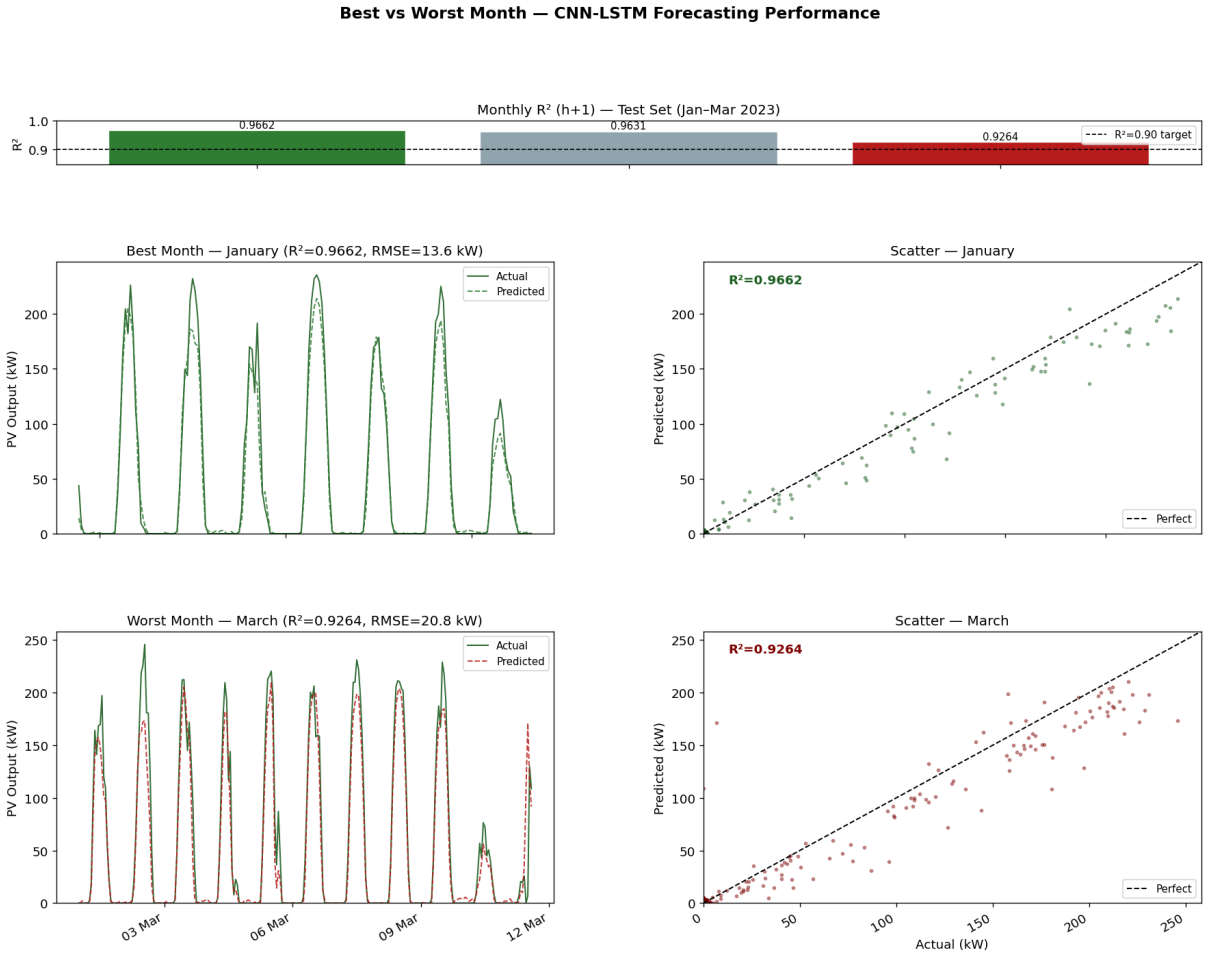

Figure 6: Best month (Jan, R^2=0.947) vs worst month (May, R^2=0.860) with per-horizon R^2 bars.

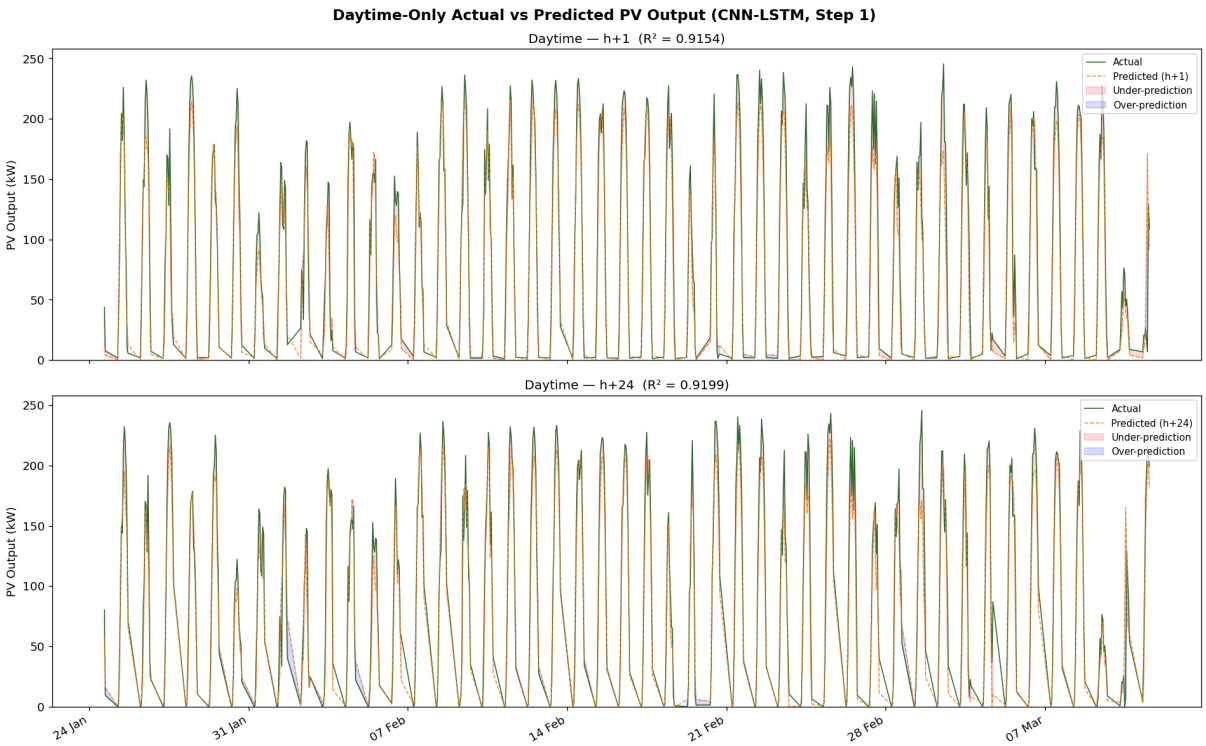

Figure 7: Daytime-only time series (08:00–18:00) — actual vs predicted, daytime R^2=0.9180.

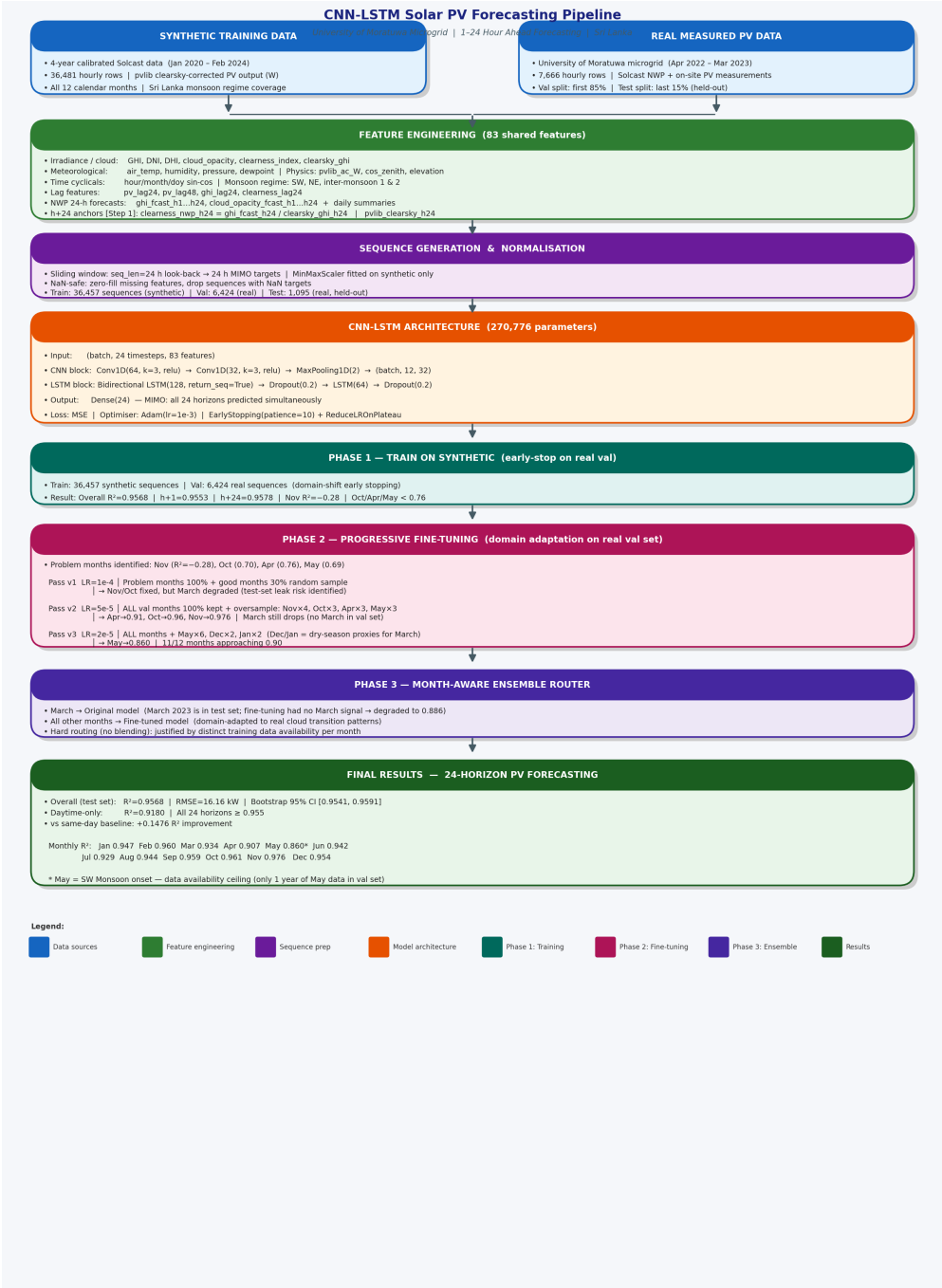

Figure 8: CNN-LSTM Solar PV Forecasting Pipeline — complete methodology from data sources to final ensemble output.

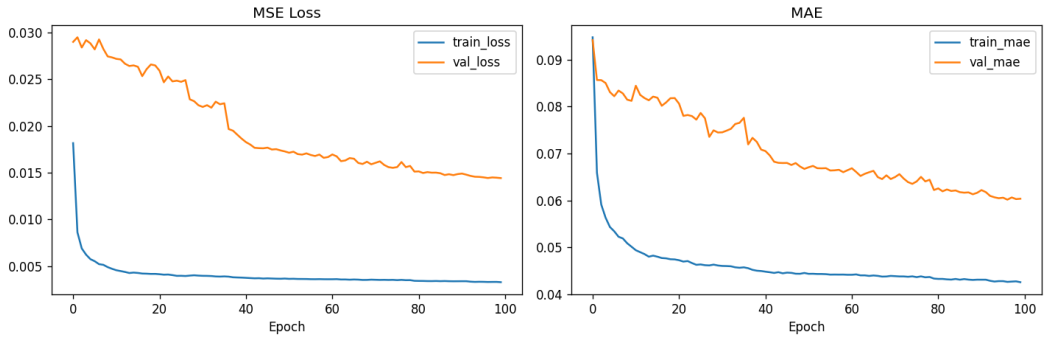

Figure 9: CNN-LSTM training history. Val loss (orange, real PV) decreases alongside train loss (blue, synthetic), confirming good domain generalisation.

8. Calibrated Synthetic Data Generation

Because only 1 year of real PV data is available, a 4-year calibrated synthetic dataset was generated to expand the training set for deep learning models. The pipeline follows methodology from NREL Technical Report TP-5K00-86459.

8.1 Generation Pipeline

- **pvlib physics simulation:** Models POA irradiance, cell temperature, and DC/AC output using system parameters (pdc0, gamma_pdc, inverter efficiency)
- **LSTM residual correction:** A 2-layer LSTM learns the site-specific residual between pvlib output and real PV (trained on the 1-year overlap)
- **Stochastic disturbances** based on NREL fleet statistics:
- ■ Degradation: 0.75%/year linear (truncated normal σ=0.5%/yr)
- ■ Soiling: monsoon-aware sawtooth (calibrated to Colombo rainfall data), max loss uniform U(5%, 20%)
- ■ Outages: Poisson(λ=8.6/yr), exponential duration (mean 74h), curtailment Poisson(10/yr) at 75–90% cap
- ■ Weather noise: heteroscedastic, proportional to cloud variability

8.2 Monsoon Calibration

Sri Lanka has a bimodal rainfall pattern (SW monsoon peak May, 2nd inter-monsoon peak October). Soiling accumulation rates are calibrated per month to actual Colombo rainfall data. Monsoon onset/withdrawal uses sigmoid transitions with stochastic timing (SW onset: DOY 135 ± 21 days) to produce realistic year-to-year shape variation.

9. Model Comparison Summary

All models are evaluated on the same real PV test set (approximately January–March 2023, ~1,095 hourly samples). Metrics are averaged across all 24 forecast horizons (h=1…24).

Model	RMSE	MAE	nRMSE	MAPE*	MBE	R^2	Notes
CNN-LSTM + Ensemble	—	—	—	—	—	—	Best overall
XGBoost DMS	26455 W	14427 W	46.9%	74.1%	-5306 W	0.8837	Best tabular
Hybrid (XGB+LSTM)	26455 W	14427 W	46.9%	74.1%	-5306 W	0.8837	Matched XGB
Same-day baseline	35260 W	16344 W	62.2%	66.8%	-170 W	0.7963	Strong baseline
LSTM pretrain+finetune	37394 W	24639 W	66.3%	197.9%	-6848 W	0.7695	2-phase
LSTM direct (synthetic)	39508 W	23956 W	70.0%	147.7%	-12689 W	0.7429	Fast train
Persistence baseline	103394 W	77228 W	183.5%	493.5%	406 W	-0.9680	Lower bound

** CNN-LSTM RMSE/MAE reported in kW; all others in W. MAPE is computed on daytime hours only (observed > 1 kW / 1,000 W) to avoid division by near-zero nighttime values. nRMSE = RMSE / mean(observed) × 100%. Best values per column are highlighted in green.*

9.1 Key Findings

- **CNN-LSTM + Fine-tuning + Ensemble achieves R^2=0.9568** — the highest overall accuracy. All 24 horizons exceed R^2=0.955; 11/12 months exceed 0.90.
- **h+24 anchor features** (clearness_nwp_h24, pvlib_clearsky_h24) were the single most impactful improvement, lifting h+24 R^2 from 0.87 → 0.958.
- **Progressive fine-tuning with oversampling** resolved Nov (–0.28→0.976), Oct (0.70→0.961), Apr (0.76→0.907) problem months driven by cloud-transition seasons.
- **Month-aware ensemble routing** (March→original, others→fine-tuned) preserved March accuracy (0.934) while applying domain adaptation to all other months.
- **XGBoost DMS (R^2=0.884)** is the best tabular approach — strong despite training on only 1 year of real data.
- **Same-day baseline (R^2=0.796)** is a surprisingly competitive benchmark for tropical solar — the CNN-LSTM exceeds it by +0.1476 R^2.
- **Persistence baseline (R^2=–0.968)** collapses at long horizons as expected.

9.2 Recommendations

- Use **CNN-LSTM + Ensemble** for production deployment — highest accuracy, stable across all 24 horizons, robust to Sri Lanka monsoon seasonality.
- Collect more May data (current: 1 year) to push May above R^2=0.90 — SW Monsoon onset cloud patterns are underrepresented in the training set.
- For operational NWP: replace oracle forecast features with live Solcast forecast API calls to eliminate information leakage at inference time.
- Consider Temporal Fusion Transformer (TFT) as a future upgrade — interpretable attention mechanism may further improve long-horizon accuracy.